



A hybrid quantum-neural network for heart disease classification

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ABSTRACT

Quantum Machine Learning represents a promising approach to address complex computational challenges beyond classical computing capabilities. In this study, we investigate its application to healthcare, focusing specifically on the classification of cardiovascular diseases. Classical Machine Learning methods often struggle with high-dimensional data, resulting in limited accuracy and reduced generalization, especially in real-world medical scenarios. To overcome these issues, we propose a Hybrid Quantum-Neural Network that combines classical Autoencoders for dimensionality reduction with Quantum Neural Networks, facilitating effective data processing compatible with current NISQ hardware. Evaluated on the Cleveland dataset, our HQNN achieves state-of-the-art classification accuracy up to 90.98%, outperforming purely classical counterparts by capturing complex feature relationships in the quantum domain. We further demonstrate the effectiveness of the proposed HQNN model in scenarios characterized by limited training data and realistic noisy quantum environments, performing robustness analyses and simulations using IBM's quantum hardware framework. These findings highlight the practical advantages of our Hybrid Quantum-Neural Network, underscoring its suitability and immediate applicability for real-world healthcare scenarios. The code is made publicly available.

1. Introduction

Quantum Machine Learning (QML) is a promising approach for solving complex computational problems beyond the capabilities of classical computers [1,2]. It has been applied in many fields seeking a quantum advantage, i.e., from finance [3] to energy [4], from remote sensing [5] to cybersecurity [6]. Healthcare is a crucial field for QML applications, due to its potential in handling large-scale and high-dimensional datasets efficiently [7]. QML could thus enhance the accuracy and efficiency of medical diagnoses, addressing the computational challenges faced by classical architectures [8].

One of the most critical areas of healthcare is the diagnosis and treatment of cardiovascular diseases (CVDs), which are the leading cause of death worldwide [9]. Findings from [10], identified nine modifiable risk factors responsible for 90% of first myocardial infarctions: smoking, dyslipidemia, hypertension, diabetes, abdominal obesity, psychosocial factors, fruit and vegetable intake, alcohol consumption, and physical inactivity. Notably, smoking alone accounted for 36% of the population-attributable risk of myocardial infarctions. In addition to their impact on mortality, CVDs also impose significant economic burdens, costing the EU economy approximately 220 billion euros yearly [11]. In light of this, prevention and management of CVDs play a fundamental role in reducing their incidence and impact on society.

Building medical datasets usually presents significant challenges, resulting in datasets with very few samples due to factors such as patient privacy, ethical considerations, and the high cost of data collection and labeling. These limitations further complicate the application of classical Machine Learning (ML) methods in this field, thus paving the way for quantum approaches. In fact, Quantum Neural Networks (QNNs) hold the potential to revolutionize computational tasks thanks to their capability to generalize faster and with fewer training samples [12–14].

Classical ML techniques have been extensively studied in this field, allowing for more accurate diagnosis and more efficient management of healthcare data [15,16]. For instance, in [17] various ML models, including Neural Network (NN), Support Vector Machine (SVM), and Decision Trees, were evaluated for heart disease classification: while the NN model showed the best performance, it failed to address feature redundancy. The same base models were tested in [18], yielding similar results; in addition, a non-traditional dataset partitioning approach was adopted, by dividing the dataset into roughly equal portions for the training and testing sets. Therefore, comparing their findings to those of other studies may not be straightforward, since their approach is designed to work with a different training-inference mechanism. Several classification algorithms for CVDs are also tested in [19], including feature extraction techniques and k-folding, while in [20] a ML method

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for cardiac disease classification was implemented with a preprocessing phase mainly focused on an accurate selection of features. Ensemble Learning has also shown great effectiveness in CVD prediction [21,22]. Bagging classifiers with majority voting were proposed in different flavors in [23–25], combining the predictive power of various base models to capture as much variance as possible from the data and achieve superior performances.

As mentioned before, there are still some limitations in using classical ML methods for the diagnosis and treatment of CVDs, as well as for high-dimensional and complex datasets. They may contain redundant or highly correlated features that can negatively impact the model's performance, as reported in [17]. Moreover, on smaller datasets, achieving satisfactory accuracy levels above 90% remains a hurdle for classical algorithms [15].

So far, few studies have addressed heart disease classification in the quantum domain. Among them, one of the first investigations to compare various QML models for heart disease classification was proposed in [26], finding that Quantum K-Nearest Neighbor (QKNN), Quantum Gaussian Naive Bayes (QGNB), Quantum Decision Tree (QDT), and Quantum Random Forest (QRF) achieved similar or even better results than their classical counterparts. A Bagging Quantum Support Vector Machine (QSVM) with Recursive Features Elimination (RFE) technique was proposed in [27], obtaining good performance on the widely used Cleveland dataset from the UCI repository for heart disease classification [28], achieving an 88.52% accuracy on the test set with the QSVM model only, increased to 90.16% with Bagging.

A hybrid model that combines improved feature extraction with an Improved Quantum Convolutional Neural Network (IQCNN) was proposed in [29], reaching satisfactory classification accuracies on several heart disease datasets. Nonetheless, a fair comparison with that approach cannot be directly discussed, since it is not clear whether binary classification or multiclass classification was considered, and how the train-test split was performed.

A quantum-enhanced ensemble machine learning approach, named QuEML, was introduced in [30]. The work compares several classical and quantum variants on a dataset from Kaggle and reports modest improvements in accuracy and training time. However, the generalization of the model under noise or in low-data regimes is not evaluated. A quantum circuit implementation of K-means clustering for unsupervised heart disease detection is implemented in [31]. While innovative in applying a quantum clustering method, their model does not consider noisy execution environments.

In this work, we present a novel Hybrid Quantum-Neural Network model for the classification of CVDs to improve the efficiency and accuracy of the related diagnoses. While other hybrid quantum models have been proposed and applied in various contexts, such as image classification [32,33], remote sensing [5,34], and time series forecasting [4,35], the approach proposed in our work introduces several key distinctions. In particular, our architecture leverages a classical Autoencoder (AE) for feature embedding, followed by a fully-quantum neural network applied to the latent representation. This architectural separation ensures that the quantum component processes a compact, information-rich representation of the input data, which both reduces dimensionality and optimizes scalability with current Noisy Intermediate-Scale Quantum (NISQ) devices.

Existing HQNN architectures often integrate quantum and classical components at each layer, forming mixed models in which classical and quantum modules alternate throughout the network [36,37]. While this design can be useful, it may introduce a computational bottleneck, limiting the degree to which the quantum model can explore high-dimensional Hilbert spaces independently. By contrast, our framework shifts the computational burden to the quantum module after classical compression, allowing the quantum circuit to operate as the sole decision-making component in the classification pipeline. This separation helps highlight the contribution of the quantum model itself and avoids limiting classical processing.

Table 1
Statement of significance.

Problem or issue	Classical Machine Learning techniques for diagnosis and treatment of cardiovascular diseases face limitations in handling high-dimensional datasets as well as in achieving satisfactory performance on small datasets.
What is already known	Neural Networks in the framework of Deep Learning hold the potential to revolutionize computational tasks in biomedical informatics.
What this paper adds	We introduce for the first time a novel Hybrid Quantum-Classical Neural Network model for cardiovascular disease classification. Quantum Neural Networks hold the potential to revolutionize biomedical computational tasks, thanks to their capability to generalize faster and with fewer training samples.

The proposed approach is designed to deal with high-dimensional datasets using current NISQ devices. This makes the proposed methodology a viable and scalable alternative to actual quantum devices, which will be proven to achieve, to the best of our knowledge, state-of-the-art performances among all comparable models and classical techniques in the literature. In order to assess the performances of the proposed approach, the Cleveland dataset was taken as a benchmark since it is widely used in the field of ML and data science to evaluate algorithms and approaches for heart disease diagnosis and prognosis. We will address a binary classification task: the goal is to correctly classify patients as 'healthy' or 'sick' based on their given features.

The rationale in using a QNN is related to its superior expressivity compared to the classical alternatives [38], especially in a context of few training data [39]. Key contributions of the present work can be summarized in the following items:

- Hybrid quantum-classical model for heart disease classification, demonstrating state-of-the-art accuracy on the Cleveland dataset, surpassing both classical and previous quantum methods with comparable model complexity.
- Dimensionality reduction via autoencoders, optimizing feature representation for quantum processing, mitigating the qubit limitations of current quantum hardware.
- Robustness in low-data scenarios, showing improved generalization compared to classical models in limited-data settings, a common challenge in medical applications.
- Validation on a realistic noisy quantum environment, using IBM's quantum hardware framework, confirming the model's feasibility under practical quantum computing constraints.¹

The rest of the paper is organized as follows. In Section 2, we illustrate the material and methods related to the proposed approach based on a Hybrid Quantum-Neural Network model. The adopted experimental setup for assessing the validity of the proposed quantum neural network model is illustrated in Section 3, while the obtained numerical results are reported in Section 4. A specific discussion about the empirical evidence from the experiments is made in Section 5, while our final general conclusions are drawn in Section 6. Finally, the Statement of Significance of this paper is summarized in Table 1.

2. Hybrid Quantum-Neural Network

The original contribution of this work is based on the use of a novel hybrid quantum-classical approach for CVDs binary classification. As illustrated in Fig. 1, the proposed Hybrid Quantum-Neural

¹ The code of the proposed system is publicly available at <https://github.com/alessioverdone/Quantum-Neural-Network-for-Heart-Disease-Classification.git>.

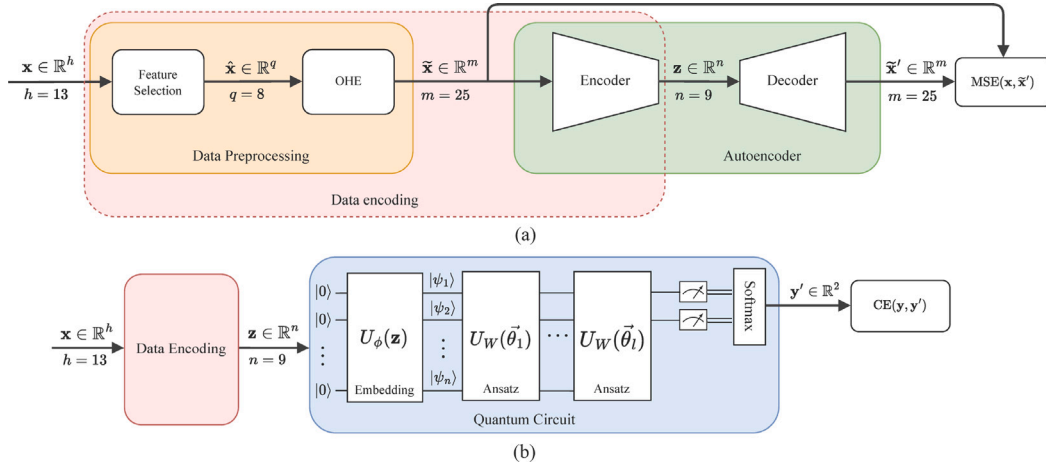


Fig. 1. Proposed HQNN framework: (a) raw healthcare data $\mathbf{x}$ are preprocessed by feature selection and ‘one-hot’ encoding (OHE), then an AE is trained on $\tilde{\mathbf{x}}$ to generate an expressive representation $\mathbf{z}$ with reduced dimensionality of the original input; (b) after classical encoding, $\mathbf{z}$ data are fed into a quantum circuit with l ansatz layers and the first two qubits are measured.

Network (HQNN) model relies on the interplay between classical AEs and QNNs. As a classical (non-quantum) encoder, it compresses inherently high-dimensional data for NISQ compatibility, while the quantum component operates in a complex Hilbert space with a reduced amount of features, and therefore qubits, for enhancing classification efficiency. The proposed solution introduces an innovative combination of classical and quantum techniques working on making quantum processing feasible, through data compression, despite the resource constraints of current quantum hardware: in this way we are able to reduce the number of required qubit maintaining at the same time high performance.

The proposed HQNN model adopts two distinct stages to effectively address the heart disease classification task. Initially, a preprocessing on raw input data $\mathbf{x} \in \mathbb{R}^h$ is performed, passing through a representation $\hat{\mathbf{x}} \in \mathbb{R}^q$ after feature selection and then to $\tilde{\mathbf{x}} \in \mathbb{R}^m$ after ‘one-hot’ encoding (OHE), where $h = 13$, $q = 8$ and $m = 25$ for the considered use case on the Cleveland dataset are respectively the input, hidden and one-hot dimension. By means of symmetrical fully connected layers, a classical AE is trained, using a Mean Squared Error (MSE) loss function, on $\tilde{\mathbf{x}}$ to capture a latent representation $\mathbf{z} \in \mathbb{R}^n$ of original input data, as shown in Fig. 1a, where n is chosen equal to 9 in order to reduce the dimensionality for the present case: the compression of the input feature dimension, by preprocessing feature selection first and autoencoder then, is essential to keep the number of input features, to each of which a qubit is associated, low and computationally efficient. This process is crucial for enhancing data quality and making feasible the subsequent processing by QNNs on currently available NISQ hardware.

The QNN operates on the latent representation $\mathbf{z}$ using variational quantum circuits (VQCs), as illustrated in Fig. 1b. QNN includes quantum data encoding, an expressive ansatz applied to the quantum state, and measurement operations. To encode classical latent data $\mathbf{z}$ into an n -qubit quantum circuit, a quantum feature map $\phi : \mathbb{R}^n \rightarrow H^{2^n}$ is implemented by applying a unitary operator $U_\phi(\mathbf{z})$ to the initial $|0\rangle^{\otimes n}$ state:

$$U_\phi(\mathbf{z})|0\rangle^{\otimes n} = |\phi(\mathbf{z})\rangle = |\psi\rangle. \quad (1)$$

A robust data encoding strategy is crucial for achieving a practical quantum advantage [40], and the choice of unitary $U_\phi(\mathbf{z})$ significantly impacts the performance of the underlying QNN. While various encoding methods exist [41,42], angle encoding using parameterized rotation gates is widely favored for its efficiency in encoding continuous variables into quantum states [38,40,43]. This method employs rotation gates to map each input feature into a qubit, allowing n input features to be represented using n qubits. In this study, we utilize a layer of parameterized R_x gates for encoding, as illustrated in Fig. 2a

Following the data encoding, an ansatz $U_W(\vec{\theta})$ of some $\vec{\theta}$ -parameterized unitaries is repeatedly applied to the state $|\psi\rangle$. Such unitaries are made of tunable single-qubit rotation gates, which are randomly initialized, along with fixed entangling gates. Rotation gates effectively manipulate the quantum state space, while entanglement facilitates the creation of highly correlated quantum vectors among all qubits. Rotation parameters are iteratively optimized utilizing a classical co-processor. To build more expressive models, rotations and entanglement layers are iteratively applied in the ansatz [44].

In this study, we chose to adopt a hardware-efficient ansatz configuration [45], which includes three alternating layers of parameterized single-qubit gates and CNOT gates with linear entanglement, as depicted in Fig. 2b. The circuit employs R_x , R_z , and R_x rotations, covering the quantum state space comprehensively and using a linear qubit connection that suits well NISQ devices. This hardware-efficient ansatz offers a good balance between circuit expressivity and trainability [43]. In the following, we set the ansatz depth at $l = 8$ in order to enhance generalization and avoid overfitting.

Measurement operations on the first two qubits in the Pauli-Z basis yield predictions via:

$$f_j(\mathbf{z}, \theta) = \langle \phi(\mathbf{z}) | U_W^{(l)}(\theta)^\dagger Z_j U_W^{(l)}(\theta) | \phi(\mathbf{z}) \rangle, \quad j = 1, 2, \quad (2)$$

where Z_j is the measurement operator on the first or second qubit, while $\theta = \{\vec{\theta}_1 \cup \dots \cup \vec{\theta}_l\}$, $U_W^{(l)}(\theta)$ represents the repeated application of the ansatz for l layers, and $f_j(\mathbf{z}, \theta)$ is the expectation on the first or second qubit state in the range $[-1, 1]$ due to the eigenvalues of the Pauli-Z basis, requiring multiple circuit shots due to quantum mechanics’ probabilistic nature. The choice of the hardware-efficient ansatz, combined with Rx-based encoding, ensures circuit expressivity while effectively controlling complexity and mitigating barren plateau effects, thus distinguishing our architecture from existing QNN-based solutions.

The outcome $f_j(\mathbf{z}, \theta)$ of the QNN is then passed to a Softmax layer, yielding a two-element vector $\mathbf{y}'$ containing the probabilistic representation of the two classes. Finally, Cross Entropy (CE) loss is computed with respect to the real targets $\mathbf{y}$ and classical optimization is used to update the whole set of parameters $\vec{\theta}_1, \dots, \vec{\theta}_l$. To update this set and train the QNN we will use the parameter-shift rule, which in a simplified scalar form can be written as:

$$\nabla_\theta f(\mathbf{z}, \theta) = \frac{1}{2} \left[f(\mathbf{z}, \theta + \frac{\pi}{2}) - f(\mathbf{z}, \theta - \frac{\pi}{2}) \right], \quad (3)$$

where θ is the single parameter to be optimized. The proposed framework is easily adaptable to other healthcare tasks, as its hybrid structure effectively addresses high-dimensional data challenges across various medical domains.

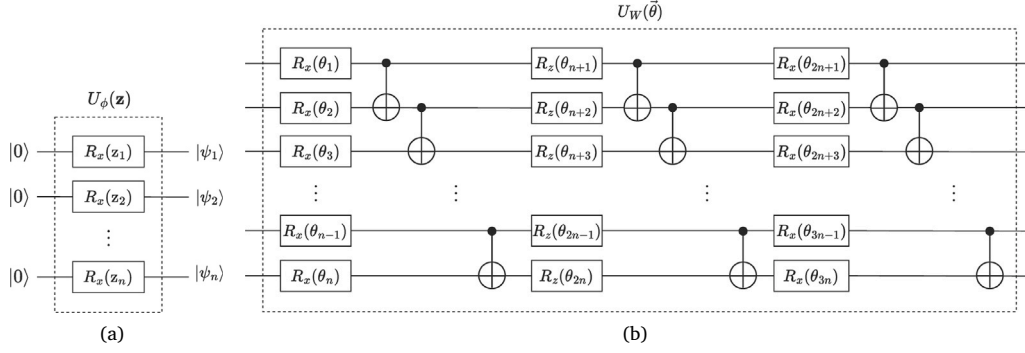


Fig. 2. Quantum circuits in the HQNN: (a) quantum feature map where each value z_i , $i = 1 \dots n$, serves as the argument of a parameterized R_x rotation gate; (b) high-level schematic representation of the proposed hardware-efficient ansatz (any level).

Table 2

Description of features in the Cleveland dataset.

Feature	Type	Description
Age	Numerical	Patient's age in years
Sex	Categorical	Patient's gender
Cp	Categorical	Type of chest pain
Trestbps	Numerical	Resting blood pressure (mm/Hg)
Chol	Numerical	Blood cholesterol level (mg/dL)
Fbs	Categorical	Fasting blood sugar level
Restecg	Categorical	Resting electrocardiogram results
Thalach	Numerical	Maximum heart rate reached (bpm)
Exang	Categorical	Exercise-induced angina
Oldpeak	Categorical	Depression of the ST segment
Slope	Categorical	Slope of ST segment during exercise
Ca	Categorical	Number of main vessels
Target	Categorical	Healthy or sick patient

3. Experimental setup

In this study, we utilized the Cleveland Heart Disease dataset [28], which contains various patient attributes aimed at predicting heart disease presence. Attributes include demographic details like age and sex, alongside physiological indicators such as blood pressure and cholesterol levels. The dataset encompasses 76 attributes, but the version most commonly referenced, including in this study, comprises 13 attributes describing the health of 303 patients with heart disease. The actual features, along with their respective domains (numerical or categorical), are shown in Table 2. Four heart disease types are categorized for binary classification. Each sample's target indicates the presence (1) or absence (0) of heart disease. The dataset has 165 (54.4%) heart disease instances and 138 (45.6%) healthy instances.

Dataset is randomly split into 80%–20% training-test sets with different seed for each run. Training data feeds the algorithm, while test data evaluates its generalization performance. Both sets are adequately preprocessed to optimize the learning process. Recursive Feature Elimination (RFE) selects key features, reducing data dimensionality and enhancing efficiency: the need to reduce the size of the original data arises from the fact that the number of input features corresponds to the number of qubits used for the quantum model, and the computational cost of utilizing a greater number of qubits increases rapidly. Its application yielded a refined set of 8 categorical features; then, categorical variables are binary-encoded for effective processing.

We assessed the HQNN's performance comparing it with both classical and quantum models. Performance is gauged using accuracy and F1-score for binary classification. As shown in Table 3, the comparison includes various classical ML methods and other quantum approaches [26,27]. We trained using ADAM optimizer with a learning rate $1 \cdot 10^{-3}$ for AE and $4 \cdot 10^{-2}$ for QNN, with 700 epochs and batch size 16 for AE, 100 epochs for QNN. As mentioned above, after preliminary

Table 3

Performance comparison of all evaluated modelson the Cleveland dataset.

Author	Model	Accuracy (%)
Meshref et al. [17]	Random forest	84.25
Classical baseline	MLP	84.92
Classical baseline	Transformer	84.09
Rabbi et al. [18]	Support vector machine	85.00
Latha et al. [24]	Majority voting	85.48
Tama et al. [22]	Two-tier classifier	85.71
Alim et al. [46]	Random forest	86.94
Mehanović et al. [23]	Majority voting	87.37
Kumar et al. [26]	Quantum random forest	89.00
Abdulsalam et al. [27]	Bagging-QSVC	90.16
Proposed method	HQNN	90.98

Table 4

Selected hyperparameters for the proposed HQNN.

Stages	Learning rate	Epochs	Batch size	Quantum layers	Qubits
QNN	$4 \cdot 10^{-2}$	100	–	8	9
AE	$1 \cdot 10^{-3}$	700	16	–	–

evaluation we found an optimal HQNN configuration consisting of $n = 9$ qubits and $l = 8$ quantum layers. The selected hyperparameters for both classical and quantum components of the proposed HQNN are summarized in Table 4.

The hyperparameters have been chosen via an extensive grid search aimed at balancing convergence stability and computational efficiency. As highlighted in Table 4, we can observe some differences in the choice of optimal hyperparameters between the AE and QNN. This is mainly due to the different nature of the classical/quantum approach. For example, the optimal learning rate for the QNN is higher than that of the AE. This is mainly due to the fact that a lower learning rate with this specific number of qubits (e.g. $n = 9$) can potentially lead to a barren plateau. At the same time, the QNN requires a lower number of epochs to converge with respect to the AE.

We have also conducted additional experiments using alternative quantum encoding strategies, including IQP encoding and Amplitude encoding, to directly assess their effectiveness on the Cleveland dataset; however, the results in terms of accuracy were unsatisfactory compared to Angle Embedding. This supports our decision to adopt R_x -based encoding, as it provided the best compromise between circuit depth and efficiency, robustness to noise, and predictive performance. In this context, the reduced circuit depth and lower parameter overhead of Angle Embedding offered a practical advantage over more expressive but less stable alternatives, making it the most suitable choice for the proposed HQNN.

For experiments, Scikit-Learn and PyTorch build classical models, while Jax and PennyLane implement QNNs. To explore quantum

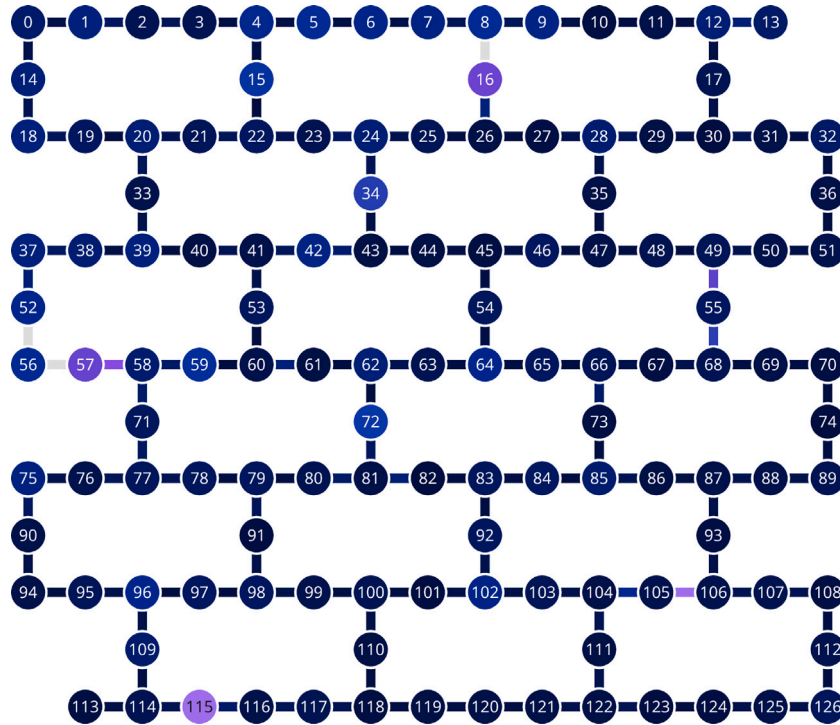


Fig. 3. Topology of IBM Sherbrooke[®] Quantum Processing Unit (QPU) [47].

hardware noise, simulations using the PennyLane-Qiskit plugin are conducted for emulating IBM Quantum systems. This includes crucial system details like coupling maps and error rates. IBM Quantum Sherbrooke[®] [47], a 127-qubit superconducting computer, is selected as the backend, and its topology is illustrated in Fig. 3.

This choice ensures that our experiments are as close as possible to the real execution environment of quantum circuits. A remark should be made about the number of circuit runs (shots). Specifically, for the noiseless simulations performed with JAX, no finite number of shots is used, as this simulator computes the expected value analytically, providing deterministic results. Conversely, for the noise simulation experiments and executions on IBM Quantum hardware, the default Qiskit configuration of 1024 shots is adopted to estimate expectation values through sampling. The simulation setup is designed to reflect key noise and topology characteristics observed in IBM's Sherbrooke[®] QPU. Among these characteristics, the coupling map plays a crucial role, as it defines the connectivity constraints between qubits and determines the number of swap operations required for implementing two-qubit gates. Gate error rates have also been considered to model the probability of execution failures due to hardware imperfections, with the median SX error rate recorded at 2.505×10^{-4} . Readout errors, which introduce inaccuracies when measuring qubit states, have also been taken into account, with a median readout error of 1.733×10^{-2} . Another critical aspect of quantum noise is decoherence, which affects the stability of quantum states and determines how long a qubit can retain its information before interactions with the environment lead to state collapse.

The IBM Quantum Sherbrooke[®] backend provides median T1 and T2 values of 284.07 μ s and 170.84 μ s, respectively, which have been integrated into our noise modeling approach. Furthermore, the impact of shot noise, an intrinsic feature of quantum measurement that contributes to output variability, has been considered in the simulations. The basis gates available in IBM Quantum Sherbrooke[®], specifically ECR, ID, RZ, SX, and X, define the set of quantum operations natively supported by the hardware and impose constraints on circuit design and execution. The IBM Quantum Sherbrooke[®] backend was selected

because it provides a high-fidelity noise model derived from real quantum hardware, ensuring that our results are not based on arbitrary assumptions but rather on empirically validated error rates. The fake provider in IBM's Qiskit framework reproduces the statistical behavior on actual hardware by incorporating data from periodic calibration reports of physical quantum devices. This means that the error rates used in our experiments match those reported from actual IBM hardware; the decoherence and gate errors reflect realistic execution scenarios rather than purely theoretical noise models.

Our chosen approach of leveraging IBM's hardware-calibrated noise models inherently accounts for a combination of various noise effects, as IBM Quantum Sherbrooke[®]'s noise profile already integrates these physical sources of errors. Furthermore, all our experiments are performed by using a commercial computer with an AMD[®] Ryzen[™] 7 5800X, an 8-core CPU with a frequency of 3.80 GHz and 64 GB of RAM.

4. Numerical results

To evaluate our HQNN model, we conduct three experiments: a comparison with state-of-the-art models for heart disease classification, demonstrating its superiority; an assessment of its performance against classical counterparts with limited training data and resources; a validation of the results on a noisy quantum backend simulator, mimicking real-world quantum computing challenges.

4.1. Classical vs. Quantum models' comparison

As summarized in Table 3, the HQNN model outperforms existing approaches on the considered Cleveland dataset, achieving a mean accuracy of 0.9098 ± 0.0132 , which is accompanied by an F1-score of 0.9138 ± 0.0130 . These values represent the average results obtained over 10 runs, where each run has been performed using a different seed for the random number generator. Notably, the standard deviation of both accuracy and F1-score is minimal, ensuring the robustness of the results. The confusion matrix in Fig. 4 illustrates strong performance in classifying both positive and negative instances, validating its effectiveness in predicting CVD outcomes. It is important to underline that the

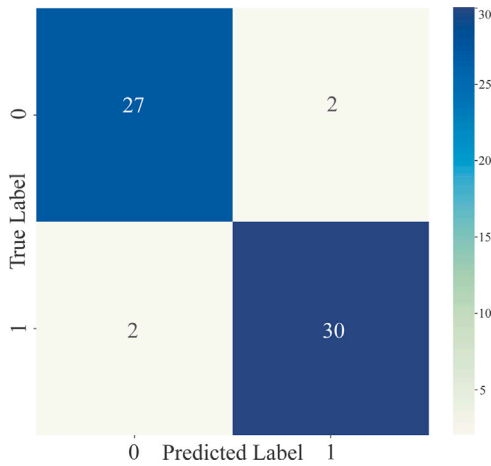


Fig. 4. Confusion matrix on the CVD classification test set of HQNN model.

confusion matrix reported in Fig. 4 refers only to a single representative run (one of the 10). This is the reason why there is a slight discrepancy between the values inferred from the matrix and the averaged metrics reported above. Moreover, we consider well-established state-of-the-art models with a similar parameter count to carry out a fair comparison.

We tried to evaluate small version of powerful models that excel in large-scale setup, such as Transformers, which, despite having significantly more parameters (approximately 8k) with respect to other models, still struggle to achieve efficient performance with limited data availability. To better understand the role of the quantum component, we performed an ablation experiment by disabling the entanglement layers in the quantum circuit. In this setting, each qubit is updated only by local single-qubit rotations, preventing the circuit from modeling feature interactions. The resulting HQNN reached a mean accuracy of 0.5623 across 10 independent runs, a dramatic decrease compared to the 0.9098 achieved with the full entangling architecture.

4.2. Classical vs. Quantum: Comparison with constrained resources

In real-world scenarios like healthcare, limited data and computational resources can hinder the effectiveness of data-driven models. With this in mind, we compare the performance of the HQNN model to a classical NN with a comparable parameter count. Specifically, our study uses a multi-layer perceptron (MLP) consisting of two stacked layers: the first with 9 neurons and the second with 18 neurons. As stated before, we choose this specific classical architecture because it has approximately the same number of parameters as the HQNN: we do this in order to ensure a fair comparison between the models. The results shown in Fig. 5 prove the models' performance across various data availability percentages and training epochs. We assess datasets from 10% to full size, with fixed epochs at 10, 30, and 60. Despite limited data and training iterations, the HQNN consistently outperforms the classical MLP in terms of both F1-score and accuracy, showcasing its reliability and robustness. For completeness, the performance of the Classical MLP, obtained with the best hyperparameters combination, is reported on the second row of Table 3.

4.3. Noisy quantum experiments

Finally, we analyze HQNN performance using an IBM fake provider module on a noisy quantum simulator. The experimental results are illustrated in Fig. 6. The optimal hyperparameters are the same reported in Table 4. The noise error rate depends on the QPU calibration and typically lies between 10^{-3} and 10^{-2} : it is determined by the hardware and therefore not controllable by the user. Moreover, we observe a

Table 5

Experimental results on further clinical datasets.

Model	Heart Dataset		SA-Heart Dataset	
	Accuracy (%)	F1-score	Accuracy (%)	F1-score
MLP	75.54	75.42	62.36	54.82
HQNN	74.45	74.23	68.81	66.59

decrease on model performances, evaluated on the same noisy model setup with 2 layers and 5 qubits, of about 5% due to the noise. Despite limited computing resources, our tests show HQNN's ability to learn meaningful representations even with few qubits, layers, and training data. Notably, training with just 10% of the Cleveland dataset yields a favorable F1-score and accuracy. Increased training periods show stable learning, but more qubits or layers do not always improve performance due to noise and complex optimization.

4.4. Other clinical datasets

In addition to the main experimental reference represented by the Cleveland dataset, we also compare classical and quantum models on other clinical datasets specifically designed for the classification of cardiac diseases. They still aim at predicting the presence or absence of heart-related conditions based on various clinical features: 'Heart' dataset contains 918 samples with 11 features; 'SA-Heart' dataset is composed of 463 samples and 10 features. It is important to point out that a systematic study on larger complex datasets is out of scope and will be addressed in future developments.

The performances for these datasets are reported in Table 5 considering two models: the baseline MLP and our proposed HQNN. The HQNN architecture consists in this case of 8 layers and utilizes 8 qubits. In contrast, the MLP has a compact architecture with 2 layers structured as (input_size, 18) and (18, classes) to maintain a comparable number of trainable parameters and ensure a fair comparison. The table clearly illustrates the comparative performance of the two models across the selected datasets: each column pair corresponds to one dataset, reporting both accuracy and F1-score.

These numerical results suggest that the proposed HQNN model performs comparably to the classical MLP when applied to datasets with a consistent number of samples. Notably, on the SA-Heart dataset, characterized by its smaller size, the HQNN outperforms the MLP, indicating that hybrid quantum models may be particularly well-suited for scenarios in which data is scarce or costly to obtain, such as in healthcare. Moreover, hardware limitations are primarily due to the use of quantum simulators rather than dedicated quantum processors. As such, we expect that real quantum hardware could further enhance performance and scalability in future implementations.

4.5. Autoencoder sensitivity analysis

We perform a sensitivity analysis by varying the autoencoder latent dimension $n \in \{2, \dots, 10\}$ and measuring both classification metrics (accuracy and F1) and complexity (training time and number of parameters) for three classifiers based on: an MLP, a Transformer, and our proposed QNN. Results are reported in Fig. 7. The best predictive performance among all models is achieved at $n = 9$. Latent dimensions smaller than 9 fail to capture sufficient feature context, leading to reductions in both accuracy and F1. In terms of model size, the QNN remains very compact (≈ 80 parameters), while the MLP and Transformer require more parameters (≈ 236 and ≈ 7906 parameters, respectively). Training time behaves differently across architectures. The QNN's training time increases sharply with n (for example, 48.8 s at $n = 9$ and 95.4 s at $n = 10$), whereas the MLP and Transformer show relatively stable and much lower timings (MLP 0.1–0.2 s, Transformer 1.6–3.7 s). These observations motivate selecting $n = 9$ as a practical trade-off: it yields peak classification performance while keeping the overall computational cost acceptable, particularly for the QNN.

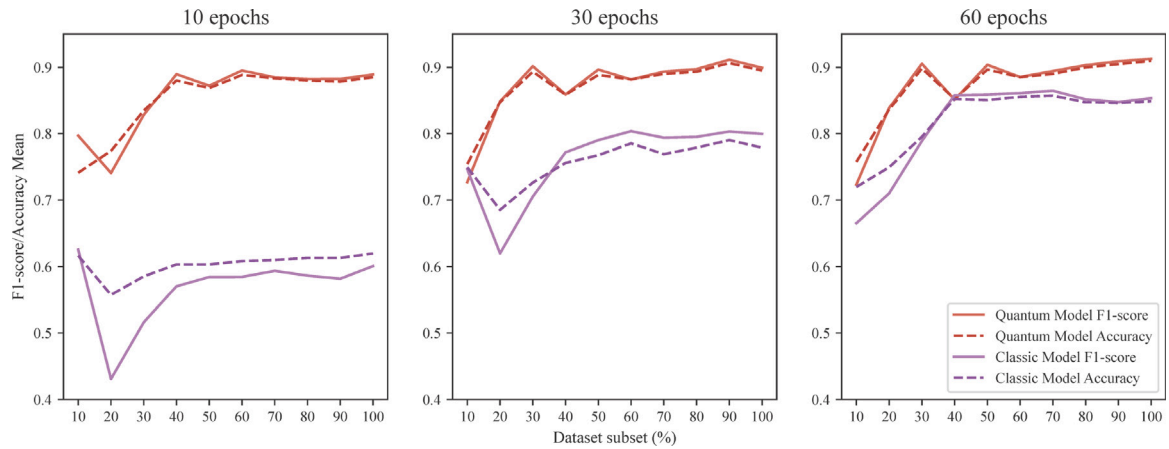


Fig. 5. Comparisons between HQNN and its classical counterpart (i.e., an MLP) with analogous number of parameters on a varying number of epochs and dataset percentages. Both models are evaluated computing accuracy and F1-score in three different training scenarios (10, 30 and 60 training epochs from left to right, respectively) and with 10 different percentages of the original dataset (from 10% to 100%).

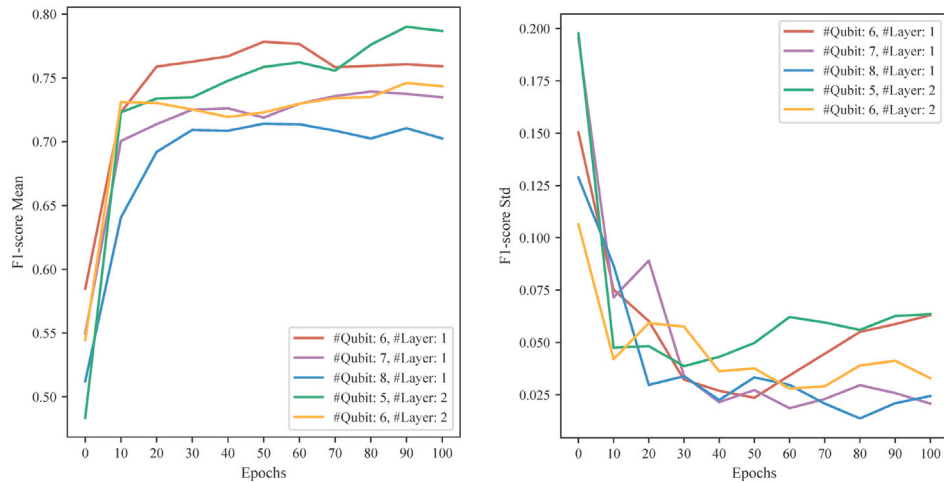


Fig. 6. HQNN tests on a noisy quantum simulator with different low-parameter configurations and limited data. Mean (left) and standard deviation (right) of the F1-score are shown at different training epochs, by training the model with only the 5% of the original Cleveland dataset.

5. Discussion

The approach carried out in this work is innovative through several perspectives: in fact, there are no quantum–classical hybrid classification models on the considered dataset in the existing literature, to the best of our knowledge. Specifically, our architecture is structured around a feature embedding phase based on a classical AE, followed by a fully-quantum NN. This design ensures that the quantum component operates in a compact, information-rich latent space derived from the original data, effectively reducing input dimensionality for NISQ devices while preserving relevant information. Many of the existing HQNN models adopt hybrid layers that interleave quantum and classical operations, which, while functional, can introduce a computational bottleneck that limits the exploitation of the quantum subspace. In contrast, our design aims to shift the computational focus to the quantum circuit, maximizing its potential expressivity and generalization capability.

This is further strengthened by the ablation experiments, which highlight the central role of entanglement in enabling the quantum model to capture joint dependencies between clinical attributes. By correlating qubits that encode different features, the circuit can represent higher-order relationships that classical feed-forward layers of similar size would require exponentially more parameters to approximate.

Thus, entanglement is the key mechanism through which the HQNN extracts clinically relevant patterns, providing both a structural and empirical explanation of the observed quantum advantage.

A key strength of the proposed HQNN lies in the contribution of the quantum component to feature extraction and generalization. By encoding the latent autoencoder output into a high-dimensional, entangled Hilbert space, the QNN can represent high-order interactions between features that classical models struggle to capture efficiently. This allows the circuit to disentangle redundancies and highlight meaningful data patterns relevant to the diagnosis. This ability also translates into superior generalization in data-constrained regimes, as observed in our few-shot training experiments, where the HQNN consistently outperformed classical networks of comparable complexity. Together, these aspects provide a clearer view of the quantum advantage in heart disease classification and reinforce the suitability of HQNNs for clinical settings where data availability is limited.

Our HQNN achieves state-of-the-art results using categorical features, with AE reducing dimensionality for NISQ readiness. Additionally, the hardware-efficient ansatz considers current quantum device limitations like qubit connectivity and error rates. As evident in Fig. 4, HQNN excels in minimizing False Negatives: this is crucial in medical diagnostics, as classical methods even in few-shot learning are generally surpassed. Furthermore, we provide empirical evidence of improved

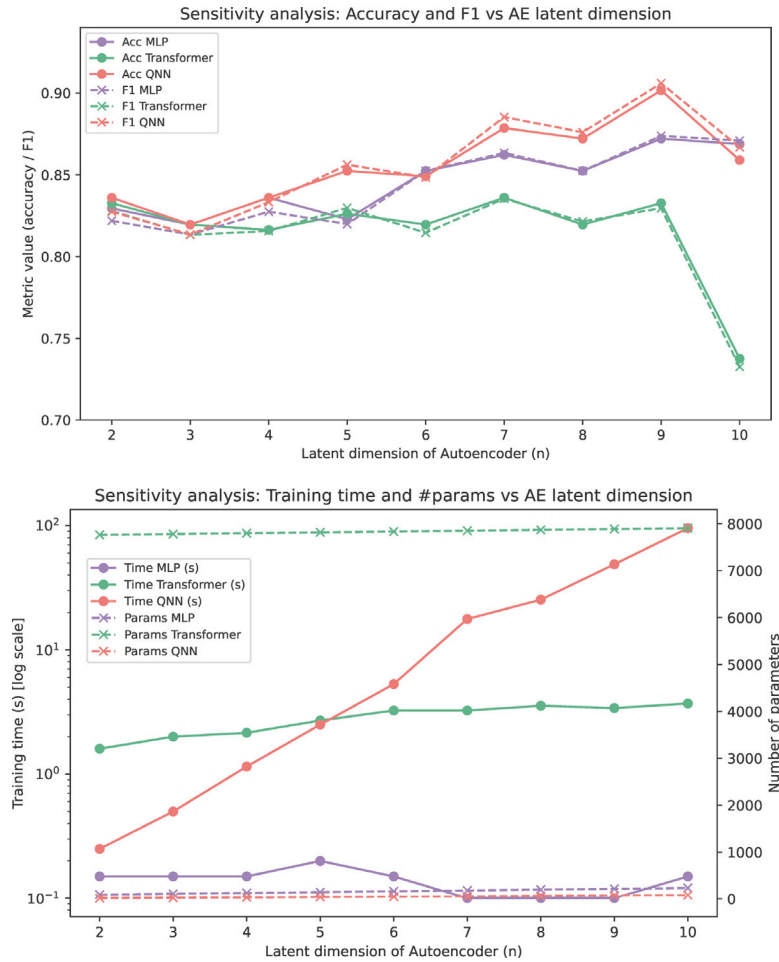


Fig. 7. Sensitivity analysis on the autoencoder latent dimension n for MLP, Transformer and QNN as n varies from 2 to 10. Top: accuracy (solid) and F1 score (dotted). Bottom: number of parameters (solid) and training time (dotted).

performance not only in terms of accuracy, but also in terms of training dynamics, showing faster convergence even in few-shot learning scenarios with limited data availability, as shown in Fig. 5. This is particularly relevant in the medical domain, where datasets are often small due to privacy, cost, or ethical considerations.

The pairwise relationship between the number of input features and qubits can be a concern when working on different datasets or scenarios; however, the non-quantum part of the hybrid model mitigates this issue. Specifically, this strategy can serve as a highly effective and practical solution from an operational standpoint. Challenges also persist in the experiments on real quantum hardware, proving costly and limiting model exploration. Despite the promising results achieved in simulation and under realistic noise models, the current generation of quantum hardware presents several limitations that constrain the practical deployment of HQNNs. NISQ devices typically offer a limited number of qubits, short coherence times, and restricted qubit connectivity, all of which impact circuit depth and expressivity. In addition, gate and measurement errors introduce instability that may hinder the reliability of medical classification tasks in real-world clinical settings. Moreover, simulators reproduce average gate and readout error rates, qubit decoherence times, and coupling constraints with high fidelity, but they do not account for calibration drifts, temporal fluctuations, and hardware access limitations that affect real devices.

However, the availability of resources is increasing over time, thanks to the public resources provided by industry players, which promote research in this way. Future advancements in quantum technologies are expected to mitigate these challenges: increased qubit

counts with higher connectivity, improved hardware fidelity, and the integration of advanced error mitigation and quantum error correction techniques will significantly expand the range of feasible QML applications [48]. For healthcare, where data is often scarce and reliability is paramount, these hardware improvements could enable HQNNs to transition from proof-of-concept studies to robust, clinically deployable tools.

Another strength of the proposed approach is its improved ability to generalize with fewer samples compared to classical models. This is particularly relevant in healthcare applications, where access to large amounts of labeled data is often limited due to privacy regulations, ethical concerns, or the high cost and complexity of data collection. In such settings, hybrid quantum models offer an appealing alternative, as they can potentially achieve robust performance even with small datasets. While current quantum hardware remains in its early stages from a practical standpoint, this study aims to showcase the promise of these techniques in real-world clinical scenarios. For example, rare disease diagnosis, personalized medicine, and early-stage clinical trials often suffer from limited data availability; in these contexts, quantum-enhanced learning could play a transformative role.

6. Conclusions

The proposed HQNN model introduces the first hybrid quantum-classical architecture with a fully-quantum data classifier for heart disease classification, achieving state-of-the-art accuracy (up to 90.98%)

and F1-score (up to 91.38%) compared to classical and quantum alternatives in the literature. The main innovation consists of combining classical Autoencoders and a quantum neural network, effectively compressing high-dimensional healthcare data into a latent space suitable for current NISQ devices. By projecting the classification task into a quantum Hilbert space, the HQNN identifies complex feature patterns not captured by classical methods, thereby enhancing predictive accuracy and efficiency. HQNN's performances are also evaluated in a simulated noisy environment emulating an IBM's actual quantum device.

Additionally, experimental results in a few-shot learning scenario demonstrate the practical advantages and scalability of the proposed approach, providing a clear step forward in hybrid quantum-classical AI for healthcare applications. Moreover, the ablation study on the autoencoder hidden dimension provides insights into the trade-off between computational cost and learning capacity of the proposed approach.

Future developments will include new supplementary experiments aimed at assessing the stability and generalization of the HQNN across diverse datasets and noise conditions. Further investigations will explore alternative ansatz architectures and optimization strategies for more efficient quantum models, as well as quantum data augmentation techniques to enhance predictive precision on limited healthcare data. Moreover, deeper analysis on larger datasets and the integration of quantum error correction techniques will be pursued to strengthen reliability and performance on real quantum hardware.

CRedit authorship contribution statement

Alessio Verdone: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation. **Federico Succetti:** Writing – original draft, Validation, Software, Investigation. **Andrea Ceschini:** Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Antonello Rosato:** Writing – review & editing, Validation, Investigation, Formal analysis. **Alessio Fioravanti:** Visualization, Supervision, Investigation. **Mas-simo Panella:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Public data are used.

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